

Milestone 4 - System Prototype Notebook Copy

Project: NLP-assisted job opportunity matching for MSBA international students

Team GitHub notebook URL:

https://github.com/Kongbai815/job-matching-nlp/blob/main/MSBA_Job_Matching_Milestone4_System_Prototype.ipynb

1. End-to-End Architecture

Input advisor query -> normalize text -> retrieve similar real postings from the 80,000-row training corpus -> apply a transparent role-fit rubric -> generate a grounded recommendation that cites source fields and flags missing authorization evidence.

2. Real Input and Output

Input query: Find entry-level US analytics or business analyst opportunities for MSBA international students with SQL, Python, Excel, Tableau, or Power BI skills. Flag cases where authorization evidence is missing.

Output headline: Review 3 of the retrieved postings first; they match analytics role-fit signals.

Rank	Posting	Company	Label
1	Entry level / Busines Data Analyst (Re	Lumos Stratgy	high_fit
2	Entry Level Data Analyst	SkiHomeRealty	high_fit
3	Entry Level Business Analyst/Data Anal	Alipro	high_fit
4	Entry Level Data Analyst - US Army (13	United States Army	unclear
5	Business Analyst in BI Analytics Guild	Swedbank	unclear

3. Evaluation

System	Rows	Accuracy	Macro F1
M2 full validation baseline	20,000	79.3%	78.5%
Same-subset hashed centroid baseline	4,000	79.7%	79.1%
M4 retrieval + grounded prototype	4,000	80.0%	80.1%

Grounding check: retrieval success rate 100.0%; support hit@5 90.0%. Authorization language is not generated unless source evidence exists.

4. Risk

The main risk is hallucinating CPT, OPT, or sponsorship eligibility. The prototype mitigates this by using template generation only and by explicitly marking authorization evidence as unavailable in the public data. The advisor remains the final reviewer.

Milestone 4 Technical Summary

System goal. Help a graduate business career advisor screen job opportunities for MSBA international students. The prototype takes an advisor query or raw posting text, retrieves similar real postings, predicts a triage label, and generates a grounded recommendation.

Architecture

Input. Advisor query or job posting text. **Retriever.** Sparse inverted index over 80,000 training postings from the 100,000-row real project sample. **Decision layer.** Transparent MSBA role-fit rubric; retrieved examples ground the answer. **Generation.** Template output that cites retrieved title, company, location, skills, label, and authorization caveat. **Output.** Recommendation plus evidence list for advisor review.

Stage	Prototype behavior
Input	advisor/student query or raw job posting text
Preprocess	normalize title, skills, location, schedule, and posting text
Retrieve	sparse inverted index over 80,000 training postings
Decide	apply a transparent role-fit rubric; retrieval supplies grounding and comparable examples
Generate	template output that cites retrieved posting fields and flags missing authorization evidence
Evaluate	accuracy, macro F1, retrieval success, and support hit@5 against weak labels

Evaluation vs. Baseline

System	Eval set	Accuracy	Macro F1
Milestone 2 published TF-IDF baseline	20k validation	79.3%	78.5%
Same-subset hashed centroid baseline	4k validation	79.7%	79.1%
Milestone 4 retrieval + grounded prototype	4k validation	80.0%	80.1%

The prototype improves same-subset macro F1 from 79.1% to 80.1%. Retrieval also provides evidence for generated output: retrieval success rate is 100.0%, and support hit@5 is 90.0%. The comparison is preliminary because the labels are weak labels, not human advisor labels.

Grounded Output Example

Input query: Find entry-level US analytics or business analyst opportunities for MSBA international students with SQL, Python, Excel, Tableau, or Power BI skills. Flag cases where authorization evidence is missing.

System headline: Review 3 of the retrieved postings first; they match analytics role-fit signals.

Entry level / Business Data Analyst (Remote) (high_fit). title/location: Entry level / Business Data Analyst (Remote) at Lumos Stratgy in San Francisco, CA; skills: sql, python, r, excel, tableau, power bi; authorization evidence is not available in this public source

Entry Level Data Analyst (high_fit). title/location: Entry Level Data Analyst at SkiHomeRealty in Anywhere; skills: sql, python, r, tableau, power bi; authorization evidence is not available in this public source

Entry Level Business Analyst/Data Analyst (high_fit). title/location: Entry Level Business Analyst/Data Analyst at Alipro in Tysons, VA; skills: sql, python, aws, excel, tableau, flow; authorization evidence is not available in this public source

Entry Level Data Analyst - US Army (13J) (unclear). title/location: Entry Level Data Analyst - US Army (13J) at United States Army in Long Beach, CA; skills: skills not listed; authorization evidence is not available in this public source

Risk and Grounding Analysis

Risk 1: hallucinated authorization claims. The dataset does not contain reliable CPT, OPT, or sponsorship text. Mitigation: template generation never claims eligibility; it explicitly says authorization evidence is missing and requires advisor review.

Risk 2: overconfidence from weak labels. High_fit, medium_fit, low_fit, and unclear are rule-derived labels. Mitigation: the output shows retrieved evidence and label reason, and the system is framed as advisor support, not an automatic forwarding or rejection tool.

Next step: add de-identified Career Center postings that include authorization language, then evaluate answer faithfulness and high-fit precision with human-reviewed labels.